A. Round Trip

time limit per test 1 second
memory limit per test 512 megabytes
input standard input
output standard output

Rating: 800 | Tags: games, greedy, implementation, math | https://codeforces.com/problemset/problem/2161/A

They say that if you do 1000 rounds, there's a secret cartoon in the end (c)

Petya and Vasya love participating in Codeforces contests. Vasya made a bet with Petya that he will take part in more rated rounds than him. Initially, Vasya's rating is R_0. There will be n rounds conducted in total, each of one of two types:

- div. 1 — rated for all participants
- div. 2 — rated for participants with rating strictly less than X, and unrated for all others,

In an unrated round, Vasya cannot change his rating. If Vasya's rating before a rated round was R, then for any **non-negative** integer x between R-D and R+D (inclusive) Vasya can adopt a strategy such that his rating becomes exactly x afterwards (here D is a positive integer). Note that rating may never become negative.

Help Vasya determine the maximum number of rated rounds he can participate in.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 1000$). The description of the test cases follows.

The first line of each test case contains four integers: R_0, X, D, n ($0 \le R_0 \le 10^9$, $1 \le X \le 10^9$, $1 \le D, n \le 1000$) — Vasya's initial rating, the rating threshold between divisions, the maximum rating delta, and the number of rounds.

The second line of each test case contains a string of size n. The string will only contain the characters "1" and "2", representing div. 1 and div. 2 rounds respectively.

The sum of n across all test cases does not exceed $3 \cdot 10^4$.

Output

For each test case, print a single integer — the maximum number of rated rounds Vasya can participate in.

Example

Input
4 2100 2100 5 3 222 2098 2100 5 6 111211 2115 2100 226 7 2211121 0 10 4 8 22111121

Output
0 6 5 8

Note

In the first example, since $R_0 \geq X$, each div. 2 round is **unrated** for Vasya, so his rating never changes. Therefore, he cannot make any round rated for himself, and the answer is 0.

In the second example, one of the optimal sequences of ratings after each round is: 2098 → 2103 → 2101 → 2099 → 2097 → 2097 → 2092.